MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PIN, RIGHT ANGLE PLUG, ELECTRICAL, POLARIZED FOR PRINTED WIRING BOARDS (.090 SPACING)

Ⓒ This specification inactive for new design after 1 October 1986

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-55302.

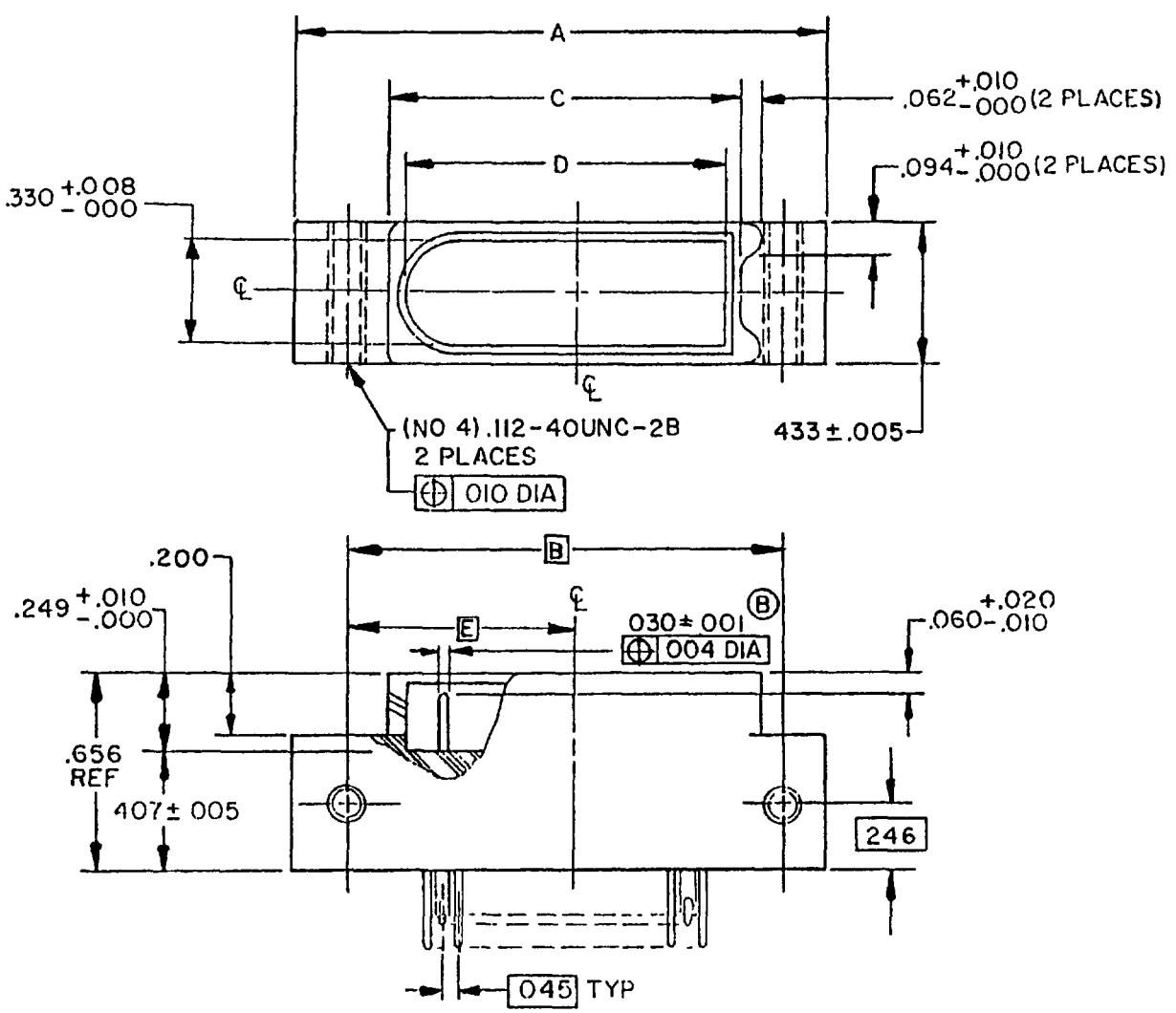
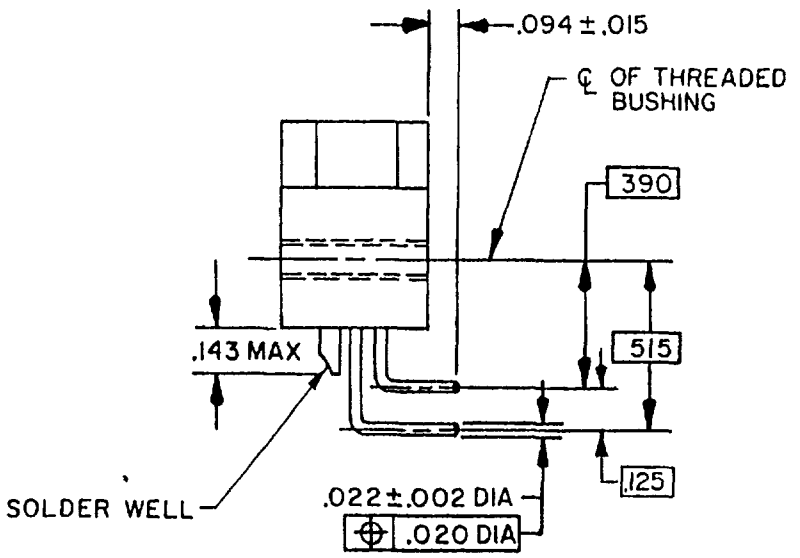


FIGURE 1. Connector, plug, .090 (2.29 mm) contact spacing, pin and solder terminals.

Ⓒ denotes changes

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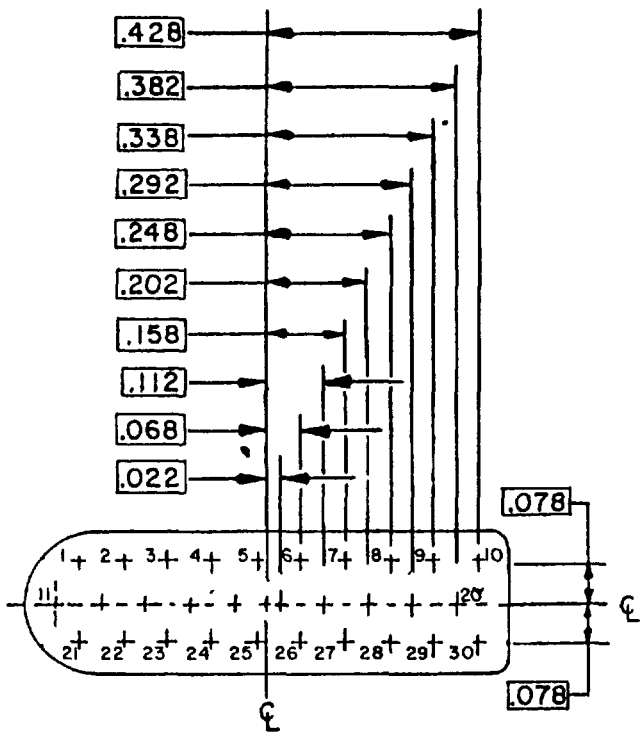


Inches	mm	Inches	mm
.002	0.05	.125	3.18
.005	0.13	.143	3.63
.008	0.20	.200	5.08
.010	0.25	.246	6.25
.015	0.38	.249	6.32
.020	0.51	.330	8.38
.022	0.56	.390	9.91
.045	1.14	.407	10.34
.060	1.52	.433	11.00
.062	1.57	.515	13.08
.094	2.39	.656	16.66
.112	2.84		

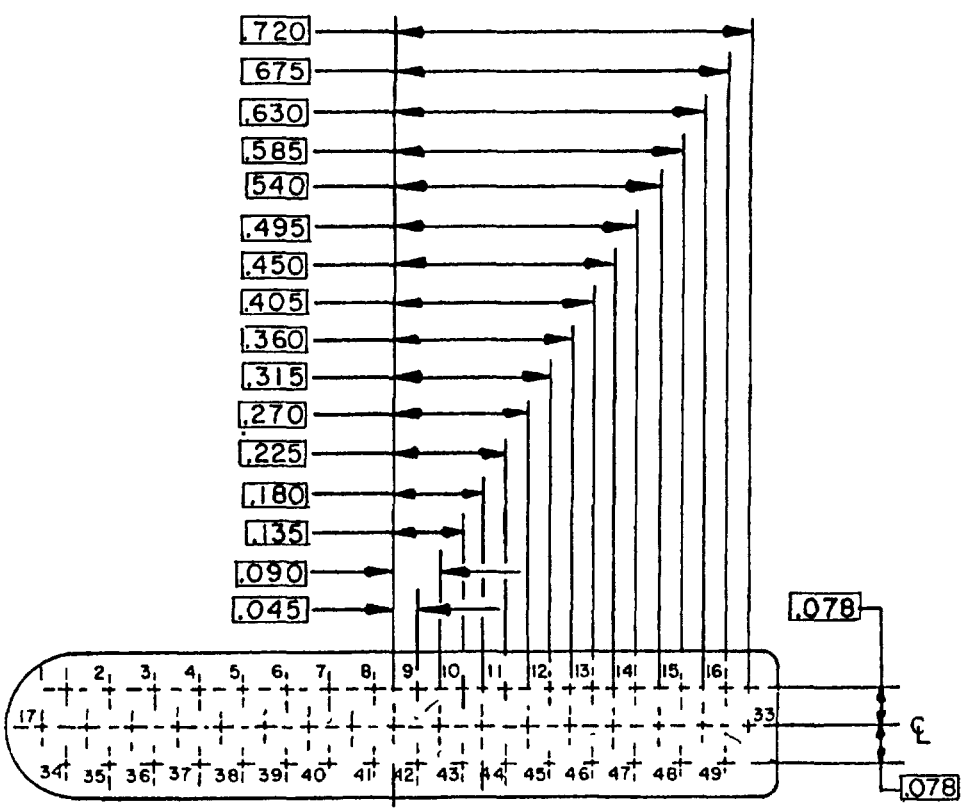
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Nominal spacing between any two adjacent pin contacts shall be .090 ±.003 (2.29 ±.08 mm).
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. For dimensions A through E see table I.

FIGURE 1. Connector, plug, .090 (2.29 mm) contact spacing pin and solder terminals - Continued.

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Inches	mm	Inches	mm
.022	0.56	.292	7.42
.045	1.14	.315	8.00
.068	1.73	.338	8.59
.078	1.98	.360	9.14
.090	2.29	.382	9.70
.112	2.84	.405	10.29
.135	3.43	.428	10.87
.158	4.01	.450	11.43
.180	4.57	.495	12.57
.202	5.13	.540	13.72
.225	5.72	.585	14.86
.248	6.30	.630	16.00
.270	6.86	.675	17.15
		.720	18.29



NOTES:
1. Contacts 1, 20, and 21 (top of page) are identified on the front face of the connector and contacts 1, 10, 11, 20, 21 and 30 are identified on the rear face per the insert arrangement.
2. Contacts 1, 16, and 49 (bottom of page) are identified on front face of the connector and contacts 1, 16, 17, 33, 34, and 49 are identified on the rear face per the insert arrangement.

FIGURE 2. Insert arrangement for .090 (2.29 mm) contact spacing connector (Male engaging face).

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-C-55302, except:

Body - reinforced epoxy resin.

Bushing - monel, QQ-N-281, class B, cold drawn.

© Plating: The contact plating shall be gold in accordance with MIL-G-45204, type II, grade C, class 1 over copper in accordance with MIL-C-14550, class 4.

Contact identification: Contact locations are identified numerically as shown on figure 2.

Mating and unmating: The maximum mating force, in pounds, shall be 0.5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: With a test current of 3 A dc, the average resistance of all contact pairs measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:

Sea level: 1,000 V rms, 60 Hz, ac.

High altitude: 275 V rms, 60 Hz, ac.

Contact current rating: 3.0 amperes.

Insert arrangement: See figure 2.

Mating connectors: See MIL-C-55302/68.

Part number: M55302/67-(dash number from table I).

TABLE I. Dash number and dimensions.

Dash number	Dimensions 1/				
	A +.010 -.000 (.25)	B	C +.010 -.000 (.25)	D +.008 -.000 (.20)	E
01	1.665 (42.29)	1.376 (34.95)	1.096 (27.84)	.996 (25.30)	.688 (17.48)
02	2.246 (57.05)	1.960 (49.78)	1.686 (42.82)	1.581 (40.16)	.980 (24.89)

1/ Metric equivalent are given for general information only. Millimeters are in parentheses.

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Custodian:
Air Force - 17

Review activities:
Air Force - 11, 17, 85
DLA - ES

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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